

Dominion and Divergence: Creighton, Econometrics, and the National Energy Program

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Abstract

We reassess the economic effects of Canada’s 1980 National Energy Program (NEP) using a balanced provincial panel from 1975–1995 and three complementary causal designs: two-way fixed-effects difference-in-differences (TWFE DiD), dynamic event studies, and ridge-regularized synthetic control (SCM). Alberta is treated as the exposed unit and compared to a data-driven counterfactual formed from other provinces.

Across specifications, Alberta’s GDP per capita after 1981 is at most modestly below its counterfactual, while compensation per worker and the compensation-to-GDP (real) ratio rise. We interpret this pattern as consistent with rent reallocation rather than structural transformation.

To complement these causal estimates, we examine the external structure of Canada’s oil sector using trade data from the Canadian International Merchandise Trade (CIMT) database. We show that crude oil exports were already overwhelmingly concentrated in the United States by the late 1980s and reached near-complete dependence by 2000.

Taken together, the results suggest that the NEP operated within a highly constrained external market environment. While policy influenced the internal distribution of economic outcomes, it did not alter the structure of the export system that shaped those outcomes.

JEL codes: C21, C23, N42, N52, Q48.

Keywords: National Energy Program; synthetic control; difference-in-differences; Canada; energy policy; provincial panel.

1 Introduction

For Canada, 1980 was a defining moment. While the Quebec referendum loomed large, an arguably more consequential event for federal–provincial relations was the announcement of the National Energy Program (NEP) in the October federal budget.¹ Introduced by the returning Liberal government, the NEP sought to reassert federal authority over energy rents through price controls, revenue reallocation, and state equity. Its stated aims—“security, opportunity, and fairness”—met a political economy already in motion: Alberta had been renegotiating resource rents and autonomy through the 1970s, even as the oil patch became more tightly linked to U.S. capital and continental markets.

This paper reads the episode through a Neo-Laurentian lens—Canada’s core–periphery dynamic in which federal instruments can centralize rents to stabilize the core—and tests its implications using modern causal designs. With a 1975–1995 province–year panel, we combine TWFE difference-in-differences, an event study, and a ridge synthetic control. Across designs, Alberta’s GDP per capita after 1981 is roughly unchanged to modestly below its counterfactual, while *compensation per worker* and the *compensation-to-GDP (real) ratio* rise. We interpret this not as a general labour–capital decoupling but as a pattern consistent with temporary rent-centralization pressures at the hinge between late-stage national planning and the continental integration later formalized by CUSFTA/NAFTA.

To complement these causal estimates, we also examine the external structure of Canada’s oil sector using trade data from the Canadian International Merchandise Trade (CIMT) database. This allows us to assess the extent to which the NEP operated within a structurally constrained export market.

¹See Government of Canada (1980a), pp. 1, 6, 9–11.

2 Historical Context

Read in isolation, Donald Creighton’s narrative can look like an east–west triumph under firm central authority. From *The Commercial Empire of the St. Lawrence* to *Dominion of the North*, he emphasizes geography, finance, and statecraft as engines of national development (Creighton, 1937, 1957). Set against that is a resource political economy in which global prices, capital, and institutions shape regional fortunes—especially in oil (Yergin, 1991; Foster, 1986).

The 1975 Syncrude rescue crystallized this politics of energy rents: after a private withdrawal, Ottawa and Ontario joined Alberta as equity partners to keep the project alive—an explicit precedent for public equity in the oil sands (Pratt, 1976). Five years later, the October 1980 Budget framed energy as national political economy—securing supply, reallocating rents, and increasing Canadian control—foreshadowing the NEP’s price, revenue, and ownership pillars (Government of Canada, 1980a).

Mini-timeline (selected waypoints).

- 1961: National Oil Policy (east–west market segmentation).
- 1973–74: First oil shock; Alberta begins major royalty/tenure changes.
- 1975: Petro-Canada founded; Syncrude rescue brings federal/provincial equity.
- 1979–80: Second oil shock; deeper U.S. capital/market ties in the patch.
- 1980: Budget announces NEP pillars (price controls, revenue sharing, state equity).
- 1989/1994: CUSFTA/NAFTA formalize continental integration, bracketing the NEP era.

3 Data and Methods

3.1 Data Sources and Construction

We use a balanced provincial panel (1975–1995). To ensure consistent macro coverage, we exclude **Prince Edward Island**,² yielding Alberta plus eight comparator provinces (BC, MB, NB, NL, NS, ON, QC, SK). Monetary variables are deflated to **1986 CAD** using province-specific CPI (Tables 18-10-0004/0005). Limited early gaps (1975) are linearly interpolated to preserve balance; robustness checks drop any constructed cells.

Core Indicators (log, 1986 CAD)

- **GDP per capita**: real GDP (all industries) / population.
- **Compensation per worker**: real *compensation of employees* (all industries) / total employment.
- *Robustness*: **Compensation-to-GDP (real ratio)** = real compensation / real GDP.³

Panel construction.

All series are pulled from Statistics Canada bulk tables (Appendix Table A3) and harmonized on (Province, Year) with consistent geography labels. Province-specific CPI (18-10-0004-01, -0005-01) is used to deflate nominal series to 1986 CAD; we backfill minimal early CPI gaps as noted in the Appendix. Outcomes are strictly positive before log transforms.

3.2 Econometric Framework

We triangulate with three complementary designs:

²PEI is excluded due to missing data in key series (CPI, GDP, employment) in early years, which prevents balanced panel estimation.

³Used as a robustness ratio; not the nominal national-accounts labour share, so values may exceed one.

1. **Difference-in-Differences (DiD)**: TWFE with Alberta $\times$ Post; province and year FE; province-clustered SEs.
2. **Event Study**: Dynamic DiD with event-time indicators (k) from five years before to nine years after the NEP; $k = -1$ is the reference; joint pre-trend test reported.
3. **Synthetic Control Method (SCM)**: *Main spec* is **constrained ridge** (non-negative, sum-to-one weights) with **bias correction**. We also report (i) unconstrained ridge (negative weights allowed; Ben-Michael, Feller & Rothstein, 2021), and (ii) NNLS variants with/without bias correction. Tuning λ via grid search minimizes pre-1981 RMSE.

3.3 Identification and Inference

Internal validity rests on (i) parallel trends (event-study pre-trend tests), (ii) good pre-period fit (SCM RMSE), and (iii) robustness to donor composition and placebo permutations. We report clustered SEs for DiD, permutation-based p -values for SCM, and sensitivity to donor leave-one-out.

Given the small number of provinces and the limited time dimension, we place particular weight on the finite-sample properties of our inference. In the spirit of MacKinnon’s contributions to robust and bootstrap-based testing, we (i) report province-clustered standard errors and joint F -tests for dynamic pre-trends in the DiD and event-study specifications, and (ii) rely on permutation-based p -values and donor re-weighting diagnostics for the SCM estimates rather than naive large-sample normal approximations.⁴ These choices are intended to keep nominal and effective test sizes closer in this small- N setting and to provide a transparent characterization of uncertainty around the estimated average treatment effects on the treated.

⁴See, for example, MacKinnon (2006); MacKinnon & Webb (2017, 2020).

4 External Market Structure and Export Dependence

4.1 Data and Method

This section examines the external market structure within which the NEP operated. While the preceding analysis identifies the causal effects of the policy on Alberta’s economic outcomes, it does not capture the structure of the export market that ultimately shaped those outcomes. Annual export values are aggregated by destination country.

We compute three summary measures of export structure: (i) the share of exports directed to the United States, (ii) the Herfindahl–Hirschman Index (HHI) of destination concentration, defined as the sum of squared export shares across all partner countries, and (iii) the combined share of the top three export destinations.

4.2 Stylized Facts

Canadian crude oil exports are characterized by extreme and persistent concentration in the U.S. market. In 1988, approximately 97.9% of exports were directed to the United States ($\text{HHI} \approx 0.96$), and by 2000 this share reached 100%, implying complete concentration ($\text{HHI} = 1.00$). Export concentration remained near-total through the oil sands expansion period, with U.S. shares above 99% as late as 2010.

More recently, concentration has declined modestly. By 2025, the U.S. share falls to approximately 89.8% ($\text{HHI} \approx 0.81$), with the top three export destinations accounting for roughly 95.8% of total exports. Despite this decline, Canada’s crude oil export structure remains highly concentrated by standard benchmarks.

4.3 Interpretation

These results indicate that Canada’s oil export structure was already highly concentrated by the late 1980s and became effectively fully dependent on the U.S. market by 2000. Subsequent decades of capital-intensive expansion did not materially diversify export destinations,

suggesting that production growth occurred within an externally anchored market structure.

This pattern implies that Canadian energy policy, including the NEP, operated within a system characterized by extreme export concentration, limiting the scope for policy to influence production incentives or alter long-run market structure.

Inspection of destination shares indicates that the modest decline in U.S. concentration after 2015 is driven by small increases in exports to a limited set of alternative markets, rather than broad-based diversification.

5 Results

5.1 Summary of Main Outcomes

Across TWFE DiD, event studies, and SCM, three robust patterns emerge in the 1981–1995 window under our preferred SCM specification (Constrained + Bias Corrected).⁵

1. **GDP per capita:** average ATT $\approx$ **-5.9%**, i.e., Alberta’s GDP per capita is moderately below its synthetic counterfactual.
2. **Compensation per worker:** average ATT $\approx$ **9.7%**, i.e., higher compensation per worker than under the counterfactual path.
3. **Compensation-to-GDP (real ratio):** average ATT $\approx$ **12.3%**, i.e., labour’s share of income rising relative to output.

Placebo-in-space/time tests place Alberta’s post-1981 gaps in the upper tail of the permutation distribution, and leave-one-out donor tests yield stable estimates.

⁵We treat the constrained, bias-corrected SCM estimates as primary and use TWFE DiD and event studies as supporting evidence, given the small number of provinces and the pre-period differences visible in the dynamic specifications.

5.2 Difference-in-Differences and Event Studies

Event-study estimates (Figures 1–3) reveal noticeable pre-period level differences between Alberta and the donor provinces, followed by post-1981 divergence that is broadly consistent with the SCM gaps. We report province-clustered standard errors and joint pre-trend F -tests, and interpret the TWFE DiD/event-study coefficients as descriptive evidence that complements, rather than replaces, the SCM estimates.

5.3 Synthetic Control Method

Figures 4–6 show good pre-period fit and clear post-1981 gaps for the main SCM specification (Constrained + Bias Corrected); the corresponding average gaps over 1981–1995 are summarized in Appendix Tables A4–A8. Donor weights appear in Tables A5–A9. Inference is based on permutation (placebo-in-space) p -values and diagnostic checks on pre-period fit, in line with our emphasis on reliable small- N inference.

5.4 Interpretation

The NEP did not reverse Alberta’s ascent; it reshaped it. Capital-intensive growth persisted, but compensation per worker and the real compensation-to-GDP ratio rose relative to counterfactuals, implying no simple decoupling of labour from output.

Interpreted in isolation, this pattern is consistent with a period of rent reallocation. However, as shown in Section 4, these effects occurred within an oil sector already characterized by extreme export concentration in the U.S. market. In this setting, the NEP can be understood as a late-stage assertion of federal dominion operating within a structurally constrained system, influencing the distribution of rents without altering the underlying market structure.

Limitations

Global oil volatility and tight monetary policy complicate causal separation; $N = 9$ provinces limits power; SCM’s linear weighting may understate heterogeneity. We therefore emphasize triangulation.

6 Discussion

When Allan MacEachen invoked “security, opportunity, and fairness,” Ottawa was responding to a political economy already in motion. Lougheed’s Alberta had been renegotiating resource rents and provincial autonomy through the 1970s. The NEP—via price controls, revenue sharing, and state equity—reasserted federal authority over energy rents.

However, the policy did not operate in a neutral market environment. As shown in Section 4, Canadian crude oil exports were already overwhelmingly concentrated in the United States by the late 1980s, approaching complete dependence by 2000. This evidence suggests that the oil sector was not merely becoming integrated into a continental system—it was already structurally anchored within one.

Within this context, the NEP should be understood as operating on the distribution of rents within Canada rather than on the underlying structure that generated those rents. With external demand concentrated in a single dominant market, the scope for policy to influence production incentives or market access was inherently limited.

This perspective helps reconcile the empirical results. While Alberta’s GDP per capita is modestly below its counterfactual, compensation per worker and the compensation-to-GDP ratio rise. Rather than indicating a broad labour–capital decoupling, these patterns are consistent with rent reallocation within a structurally constrained system.

More broadly, the results suggest that Canadian energy policy in this period was able to influence the internal distribution of economic outcomes, but not the external conditions under which those outcomes were determined. The NEP thus appears less as a turning point

in the structure of the oil economy, and more as a policy intervention whose effects were bounded by an already highly concentrated export market.

7 Conclusion

We revisit the National Energy Program using modern causal designs and complementary evidence on the structure of Canada’s oil export market. Across DiD, event-study, and SCM estimates, Alberta’s GDP per capita from 1981–1995 is at most modestly below its counterfactual, while compensation per worker and the compensation-to-GDP (real) ratio rise.

We interpret this pattern as consistent with rent reallocation rather than structural transformation. Evidence from trade data shows that Canadian crude oil exports were already overwhelmingly concentrated in the United States by the late 1980s and reached near-complete dependence by 2000.

Taken together, the results suggest that the NEP operated within a highly constrained external market environment. While policy influenced the internal distribution of economic outcomes, it did not alter the structure of the export system that ultimately shaped those outcomes.

More broadly, the findings highlight a limitation of economic policy in small, resource-dependent economies: governments may reallocate rents, but when export markets are highly concentrated, the scope for altering long-run economic trajectories is inherently limited.

Figures

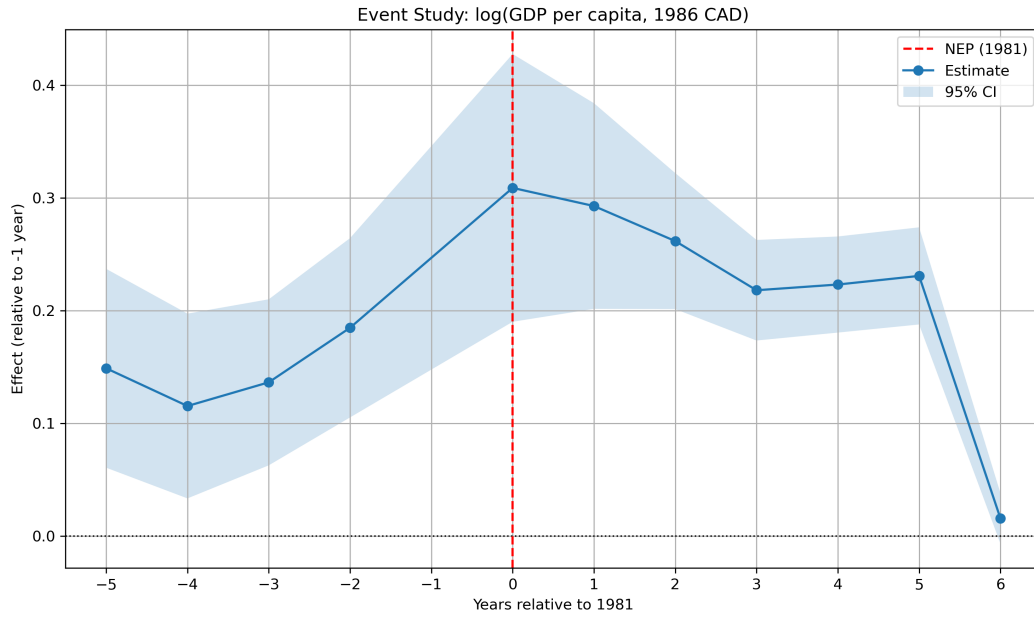


Figure 1: Event study: **log GDP per capita** (1986 CAD). Points show event-time coefficients $\hat{\beta}_k$ with confidence intervals, relative to the reference period $k = -1$ (1980). Vertical line marks $k = 0$ (1981). Dashed horizontal line denotes zero.

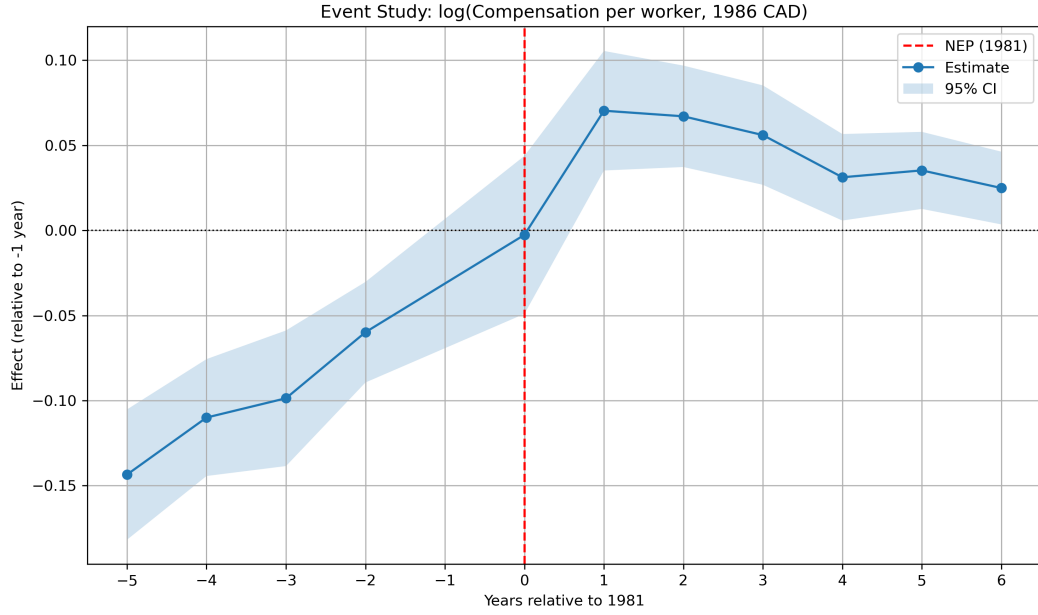


Figure 2: Event study: **log compensation per worker**. Points show event-time coefficients $\hat{\beta}_k$ with confidence intervals, relative to the reference period $k = -1$ (1980). Vertical line marks $k = 0$ (1981). Dashed horizontal line denotes zero.

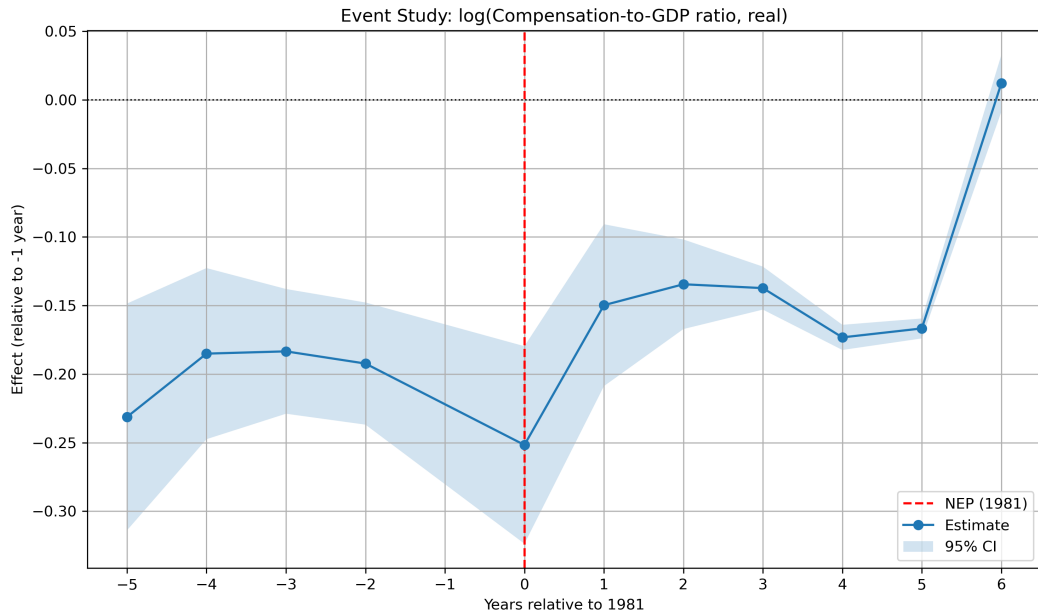


Figure 3: Event study: **Compensation-to-GDP (real ratio)**. Points show event-time coefficients $\hat{\beta}_k$ with confidence intervals, relative to the reference period $k = -1$ (1980). Vertical line marks $k = 0$ (1981). Dashed horizontal line denotes zero.

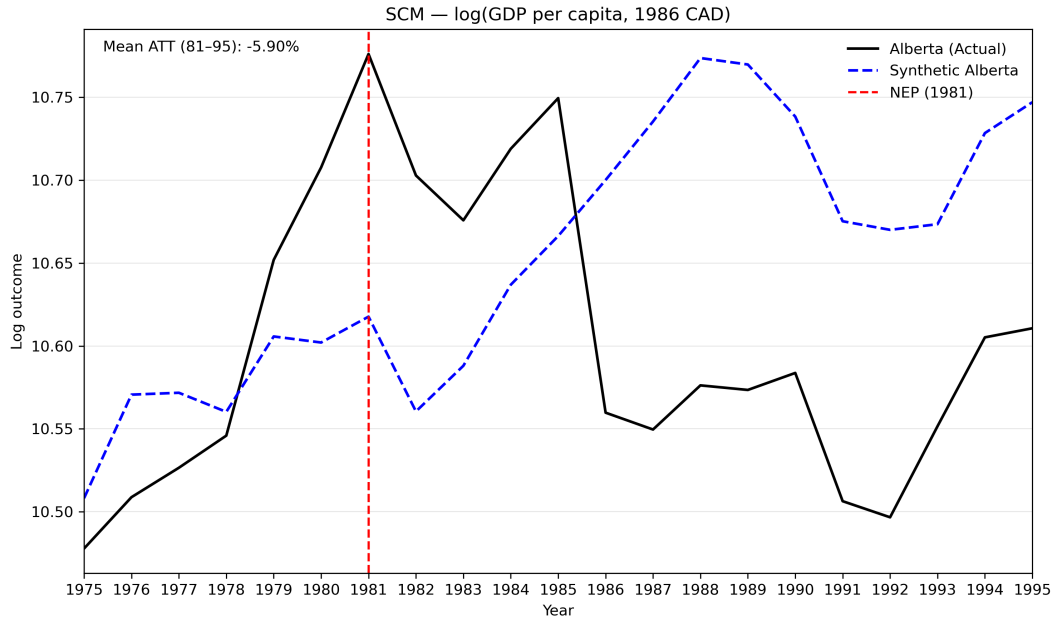


Figure 4: SCM (Main Spec): Log GDP per Capita—Alberta (solid) vs Synthetic (dashed).

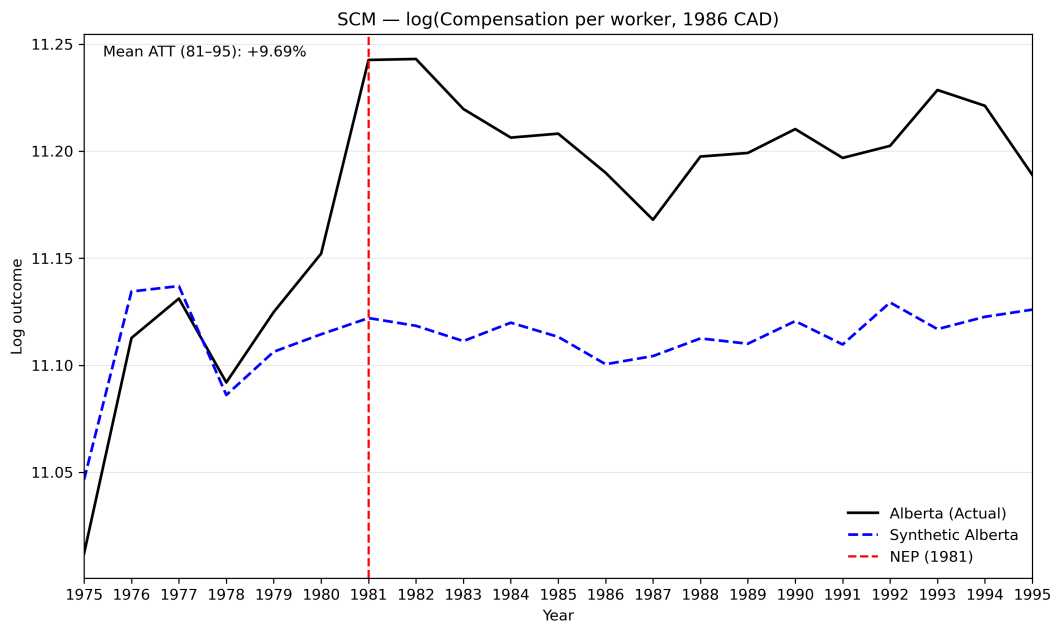


Figure 5: SCM (Main Spec): Log Compensation per Worker—Alberta (solid) vs Synthetic (dashed).

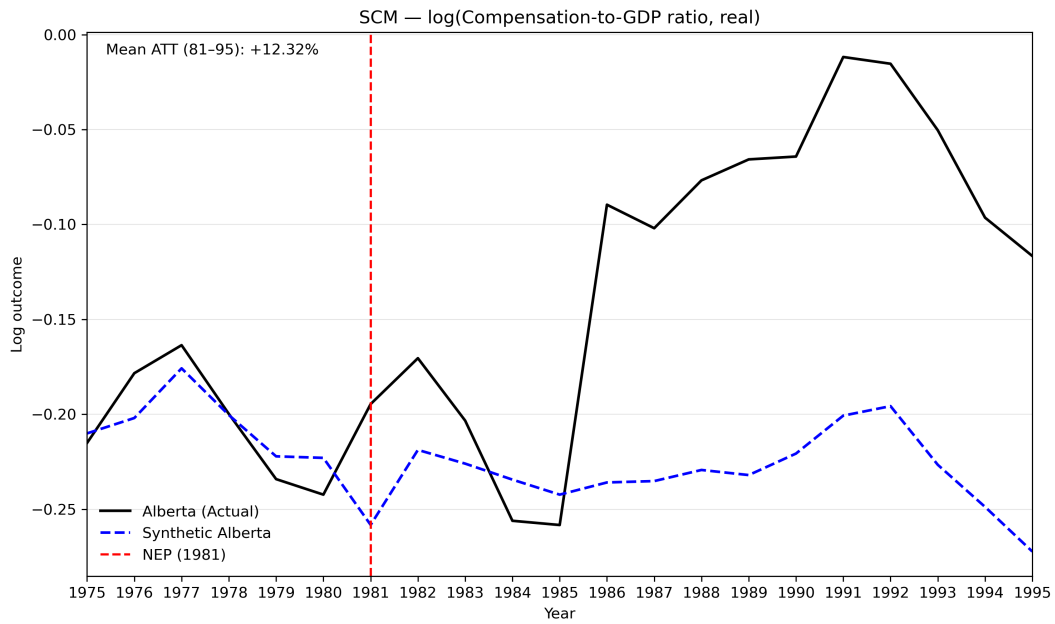


Figure 6: SCM (Main Spec): **Compensation-to-GDP (real ratio)**—Alberta (solid) vs Synthetic (dashed).

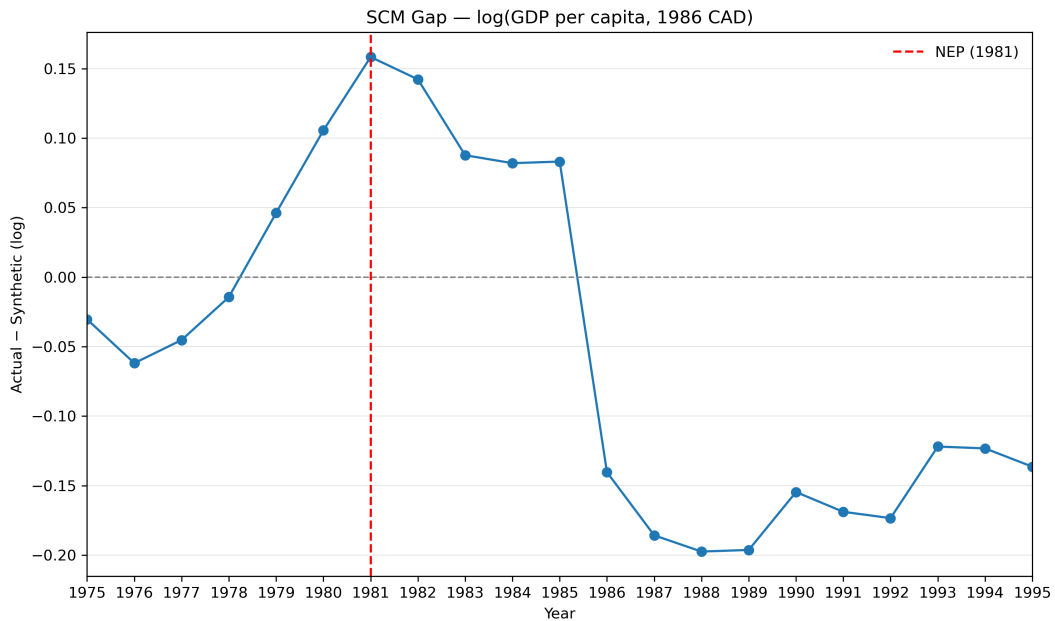


Figure 7: SCM Gap (Main Spec): Log GDP per Capita—Alberta minus Synthetic (positive $\Rightarrow$ Alberta > Synthetic).

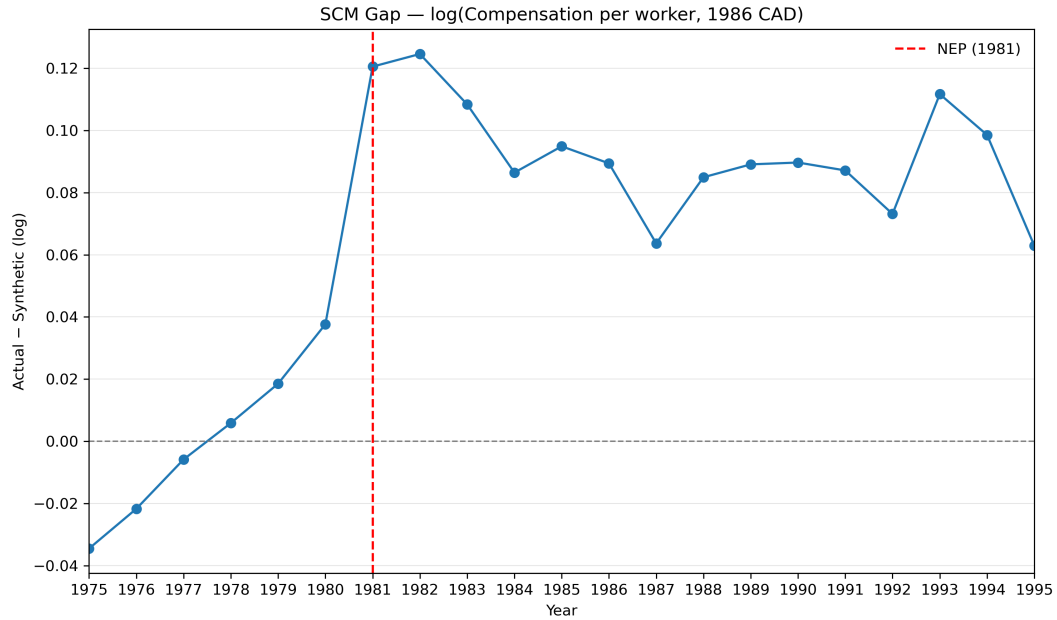


Figure 8: SCM Gap (Main Spec): Log Compensation per Worker—Alberta minus Synthetic.

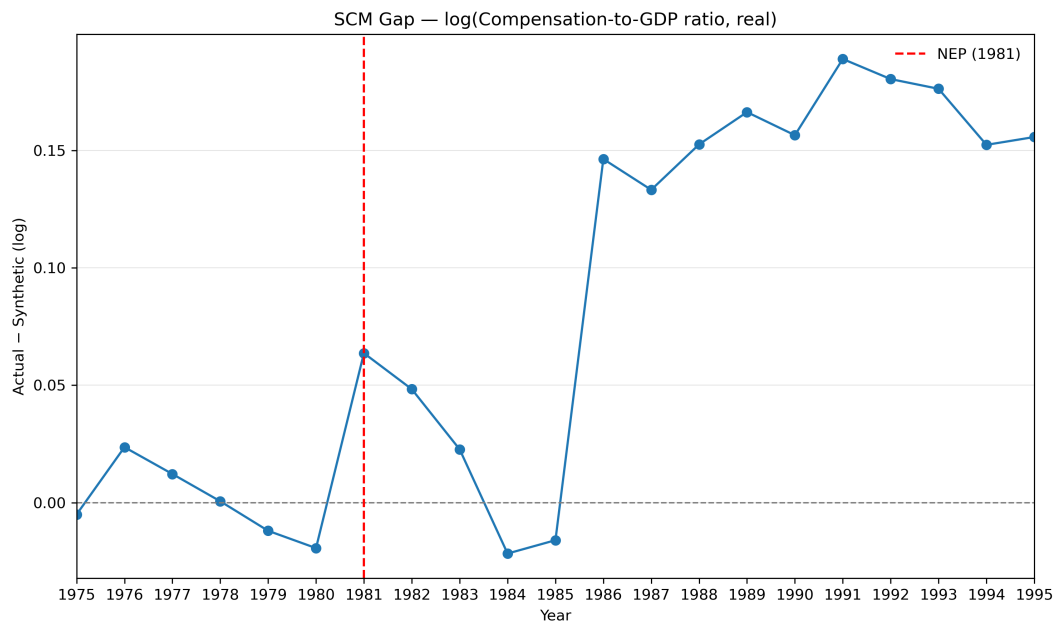


Figure 9: SCM Gap (Main Spec): **Compensation-to-GDP (real ratio)**—Alberta minus Synthetic.

Appendix: Tables

Appendix Tables Index

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A10–A12	Event Study Coefficients	Coefficients by event time k

Table A1: Summary Statistics

	N	Mean	Std. Dev.	Min	P25	Median	P75	Max
Year	189	1,985.00	6.07	1,975.00	1,980.00	1,985.00	1,990.00	1,995.00
GDP_Real_Mining	189	2,708,718,589.81	4,544,024,557.49	76,152,702.03	290,900,000.00	1,210,400,000.00	2,379,700,000.00	25,146,952,789.70
Wpw_All	189	65,440.89	7,151.42	47,079.73	60,860.66	64,451.48	71,335.72	80,795.63
Population	189	2,873,730.90	3,002,281.81	556,496.00	852,659.00	1,082,495.00	3,292,111.00	10,950,119.00
Employment	189	1,292,741.98	1,416,277.23	158,825.00	321,066.67	486,866.67	1,559,658.33	5,199,008.33
GDP_pc_real_1986	189	27,110.28	6,962.53	13,829.44	23,303.99	27,096.09	30,594.91	47,862.91
GDP_per_worker_real_1986	189	63,176.20	9,436.26	42,738.83	58,350.73	61,676.20	66,386.59	95,466.23

Notes: Monetary values in 1986 CAD; employment in persons.

Table A2: Model Overview

Design	Specification / Inference
DiD	TWFE; AB×Post; province and year FE; cluster-robust SEs
Event study	Leads/lags $k \in [-5, 9]$; $k = -1$ reference; joint pre-trend test
SCM (main)	Constrained ridge ($w \geq 0$, $\sum w = 1$) + bias correction; pre-1981 RMSE; placebos; LOO donors
SCM (alt)	Ridge with negative weights (Ben-Michael, Feller & Rothstein, 2021); NNLS $\pm$ bias correction

Table A3: Statistics Canada Web Data Service Tables Used

Hyphenated ID	PID	Description
36-10-0380-01	3610038001	GDP by industry, annual (current \$ pre-1984; also constant)
36-10-0381-01	3610038101	GDP by industry (millions of dollars; constant series)
36-10-0324-01	3610032401	Income-based GDP, provincial, 1975–1980 (deflated)
36-10-0221-01	3610022101	Income-based GDP, provincial, 1981–1995 (constant/chained preferred)
36-10-0222-01	3610022201	Compensation of employees, by industry, monthly
18-10-0004/0005	1810000401/1810000501	CPI, annual average; CPI by province
14-10-0017-01	1410001701	Labour force characteristics (employment), monthly
17-10-0005-01	1710000501	Population estimates, July 1 (all ages)
10-10-0122-01	1010012201	Financial market statistics (policy rate), monthly

Notes: Accessed via Statistics Canada WDS; all series deflated to 1986 CAD.

Table A4: SCM ATT Summary: Log GDP per Capita (Ridge and constrained specs; main spec: Constrained + Bias Corrected)

Spec	Chosen λ	Pre-RMSE (log)	Post-RMSE (log)	Mean ATT 81–95 (%)	Mean ATT 81–85 (%)	Mean ATT 86–95 (%)	Median ATT 81–95 (%)	Median ATT 81–85 (%)	Median ATT 86–95 (%)
Ridge (neg w)	0.10	0.065426	0.097691	-0.518335	13.145732	-7.389081	-5.566570	12.708350	-7.059177
Constrained	0.10	0.058915	0.142740	-7.257368	8.346449	-15.086662	-13.748344	7.945456	-15.136144
Constrained+BC	0.10	0.058915	0.142740	-7.257368	8.346449	-15.086662	-13.748344	7.945456	-15.136144
NNLS		0.050709	0.086869	-0.268693	11.837378	-6.359251	-4.982021	11.565171	-6.361064
NNLS+BC		0.050709	0.086861	-0.257126	11.850349	-6.348390	-4.971001	11.578111	-6.350204

Table A5: SCM Donor Weights: Log GDP per Capita (weights for Constrained + Bias Corrected)

Province	Weight
Saskatchewan	0.188810
British Columbia	0.168267
Quebec	0.154512
Ontario	0.143338
Newfoundland And Labrador	0.133631
Manitoba	0.119122
New Brunswick	0.057720
Nova Scotia	0.034600

Table A6: SCM ATT Summary: Compensation per Worker (Ridge and constrained specs; main spec: Constrained + Bias Corrected)

Spec	Chosen λ	Pre-RMSE (log)	Post-RMSE (log)	Mean ATT 81-95 (%)	Mean ATT 81-85 (%)	Mean ATT 86-95 (%)	Median ATT 81-95 (%)	Median ATT 81-85 (%)	Median ATT 86-95 (%)
Ridge (neg w)	0.0009	0.005846	0.103062	10.751953	10.820898	10.699627	10.841867	11.191268	10.437030
Constrained	0.0019	0.016351	0.064251	6.357113	7.136146	5.941807	6.685268	6.718042	6.443170
NNLS		0.014049	0.039951	3.364110	3.741355	3.135865	3.113195	2.772505	3.775596
NNLS (BC)		0.014049	0.039962	3.365544	3.742795	3.137296	3.114626	2.773930	3.777036

Table A7: SCM Donor Weights: Compensation per Worker (weights for Constrained + Bias Corrected)

Province	Weight
Saskatchewan	0.759970
Quebec	0.240030
British Columbia	0.000000
Manitoba	0.000000
New Brunswick	0.000000
Newfoundland And Labrador	0.000000
Nova Scotia	0.000000
Ontario	0.000000

Table A8: SCM ATT Summary: Compensation-to-GDP (real ratio) (Ridge and constrained specs; main spec: Constrained + Bias Corrected)

Spec	Chosen λ	Pre-RMSE (log)	Post-RMSE (log)	Mean ATT 81-95 (%)	Mean ATT 81-85 (%)	Mean ATT 86-95 (%)	Median ATT 81-95 (%)	Median ATT 81-85 (%)	Median ATT 86-95 (%)
Ridge (neg w)	0.10	0.026657	0.118835	9.815692	-0.762378	15.018553	12.033149	0.460968	14.806592
Constrained	0.0009	0.012076	0.127693	11.765897	2.381079	16.398589	15.169365	2.501547	16.105042
NNLS		0.028947	0.080701	5.575784	-0.144499	8.256543	6.314439	-2.409709	9.115509
NNLS (BC)		0.028566	0.083815	6.071027	0.323911	8.764362	6.813147	-1.951925	9.627357

Table A9: SCM Donor Weights: Compensation-to-GDP (real ratio) (Main: Constrained + Bias Corrected)

Province	Weight
Saskatchewan	0.503002
Newfoundland And Labrador	0.466422
British Columbia	0.030575
Manitoba	0.000000
New Brunswick	0.000000
Nova Scotia	0.000000
Ontario	0.000000
Quebec	0.000000

Table A10: Event Study: log(GDP per capita, 1986 CAD)

Variable	Estimate (s.e.)	95% CI: low	95% CI: high	p-value
Intercept	10.163 (0.002)***	10.160	10.167	0.000
NEP_bin_m5	0.149 (0.045)***	0.061	0.236	0.001
NEP_bin_m4	0.115 (0.041)***	0.034	0.197	0.006
NEP_bin_m3	0.136 (0.037)***	0.063	0.209	0.000
NEP_bin_m2	0.185 (0.040)***	0.106	0.264	0.000
NEP_bin_p0	0.309 (0.060)***	0.191	0.427	0.000
NEP_bin_p1	0.293 (0.046)***	0.202	0.383	0.000
NEP_bin_p2	0.262 (0.031)***	0.202	0.322	0.000
NEP_bin_p3	0.218 (0.023)***	0.174	0.262	0.000
NEP_bin_p4	0.223 (0.022)***	0.181	0.265	0.000
NEP_bin_p5	0.231 (0.022)***	0.188	0.274	0.000
NEP_bin_p6	0.016 (0.011)	-0.006	0.038	0.166

Notes: Province and year FE. Bin -1 omitted as baseline. Significance: * $p < .10$, ** $p < .05$, *** $p < .01$.

Table A11: Event Study: log(Compensation per worker, 1986 CAD)

Variable	Estimate (s.e.)	95% CI: low	95% CI: high	p-value
Intercept	11.084 (0.001)***	11.082	11.085	0.000
NEP_bin_m5	-0.144 (0.019)***	-0.181	-0.106	0.000
NEP_bin_m4	-0.110 (0.017)***	-0.144	-0.076	0.000
NEP_bin_m3	-0.099 (0.020)***	-0.138	-0.059	0.000
NEP_bin_m2	-0.060 (0.015)***	-0.089	-0.031	0.000
NEP_bin_p0	-0.003 (0.023)	-0.049	0.043	0.911
NEP_bin_p1	0.070 (0.018)***	0.036	0.105	0.000
NEP_bin_p2	0.067 (0.015)***	0.038	0.097	0.000
NEP_bin_p3	0.056 (0.015)***	0.027	0.085	0.000
NEP_bin_p4	0.031 (0.013)**	0.006	0.056	0.016
NEP_bin_p5	0.035 (0.011)***	0.013	0.058	0.002
NEP_bin_p6	0.025 (0.011)**	0.004	0.046	0.023

Notes: Province and year FE. Bin -1 omitted as baseline. Significance: * $p < .10$, ** $p < .05$, *** $p < .01$.

Table A12: Event Study: log(Compensation-to-GDP ratio, real)

Variable	Estimate (s.e.)	95% CI: low	95% CI: high	p-value
Intercept	0.049 (0.001)***	0.047	0.051	0.000
NEP_bin_m5	-0.231 (0.042)***	-0.313	-0.149	0.000
NEP_bin_m4	-0.185 (0.032)***	-0.247	-0.123	0.000
NEP_bin_m3	-0.183 (0.023)***	-0.228	-0.138	0.000
NEP_bin_m2	-0.192 (0.023)***	-0.236	-0.148	0.000
NEP_bin_p0	-0.252 (0.036)***	-0.323	-0.180	0.000
NEP_bin_p1	-0.150 (0.030)***	-0.208	-0.091	0.000
NEP_bin_p2	-0.134 (0.017)***	-0.167	-0.102	0.000
NEP_bin_p3	-0.137 (0.008)***	-0.153	-0.122	0.000
NEP_bin_p4	-0.173 (0.005)***	-0.182	-0.164	0.000
NEP_bin_p5	-0.167 (0.004)***	-0.174	-0.159	0.000
NEP_bin_p6	0.012 (0.010)	-0.008	0.033	0.252

Notes: Province and year FE. Bin -1 omitted as baseline. Significance: * $p < .10$, ** $p < .05$, *** $p < .01$.

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